

Batch 1 (Units I-II): Analogy + classification + number series + letter series

CONCEPT	FORMULA / RULE	TRAP
Word analogy	Name the relation, apply to pair 2 (workplace, tool, cause). Doctor:Hospital :: Teacher:School.	First-pair-only; verify both directions.
Number analogy	Operations + squares/cubes $\pm 1/2$. $7:50 = 7^2+1$, so $9:82$.	Looks-like is not proof; test reverse.
Letter analogy	A1-Z26 + shift/reverse; CAT +2 gives ECV.	A starts at 1, not 0; Z+1 wraps to A.
Figure analogy	List ALL changes (rotate, move, count); apply all to pair 2.	One-change seen, rotation missed; test extremes.
Word odd-one	Group by reason; Mango out of Rose/Lotus/Lily (fruit vs flower).	Feels-off without a reason fails.
Number odd-one	Property check; 190 out of 121/144/169 (non-square).	No-reason picks; state the property.
EXCEPT trap	EXCEPT/NOT flips the task; read twice, mark the word.	Missed EXCEPT = full fail; circle it.
First difference	Constant diffs; 3,7,12,18,25 diffs 4,5,6,7, next +8 = 33.	Stopping at first level; check diff-of-diff.
Ratio series	x/divide pattern; 3,6,12,24 x2 gives 48.	Add-only blindness when growing fast.
Alternating	Split odd/even positions; 5,2,7,4,9,6 gives odd 5,7,9,11.	Single-series assumption; split on zigzag.
Squares/primes	$n^2 \pm 1$, $n^3 \pm 1$, primes, Fibonacci sums; 2,3,5,8,13 gives 21.	+1 missed; Fibonacci-blind.
Wrong term	Find the break: 4,9,16,25,35 has 35, fix 36.	Missing vs wrong need different hunts; use options.

60-SECOND SPEED ROUND (cover answers, recall aloud)

Q: Cricket:Bat :: Tennis:?	A: Racket
Q: Next: 5,11,19,29,41?	A: Diffs 6,8,10,12, +14 = 55
Q: Opposite of J(10)?	A: $27-10 = 17 = Q$

Batch 2 (Units III-IV): Coding + blood relations + direction + ranking

CONCEPT	FORMULA / RULE	TRAP
Shift coding	+n/-n with Z-to-A wrap; CODE +2 gives EQFG.	Wrap missed; A0 miscount.
Position sums	A1-Z26 sums; BAD = 2+1+4 = 7; EJOTY = 5,10,15,20,25 anchor.	Reverse-Z mix; sums missed.
Substitution	Map every letter then apply all; ATE=BXJ (+1) gives EAT=FBV.	One-letter pattern assumed; map all.
Symbol coding	Sign map + conditions in order (vowels +2, consonants -1).	Condition order flipped.
Compare trap	CAT=24 vs BAT=22: isolate via common + difference.	Single-code solved; compare to verify.
Family tree	Draw + generations + side; maternal-uncle = mother + brother.	No tree; maternal/paternal swapped.
Coded relations	Decode fully then substitute; A+B siblings chains stay transitive.	Half decode; gap miscount.
Pointing/ONLY	Word logic; only-daughter of mother (male speaker) = sister.	Gender assumed; ONLY ignored.
Compass turns	Face first; facing East, left = North; track R/L/U turns.	Left = West reflex; turns untracked.
Path trace	N2 E3 S2 ends 3 East of start; legs cancel first.	Diagonal treated as 90 deg.
Shortest/shadow	6E+8N gives 10 NE (Pythagoras); morning shadow falls West.	Walked path counted as shortest; shadow flipped.
Rank both ends	Total = L+R-1; mirror rank = total-rank+1 (30 total, 9th left = 22nd right).	-1 missed; total taken as sum.

60-SECOND SPEED ROUND (cover answers, recall aloud)

Q: Total 30, 9th from left, rank from right?	A: $30-9+1 = 22^{\text{nd}}$
Q: Face South, turn left then right?	A: East, then South
Q: Morning sun in face, shadow?	A: Behind = West

Batch 3 (Units V-VIa): Seating arrangement + syllogisms

CONCEPT	FORMULA / RULE	TRAP
Between ranks	Between = total-L-R+overlap adjust; verify via ends sum.	Subtract-only without +2 adjust.
Interchange	Swap then recompute both ends from new positions.	Swap-blind; recompute each side.
Multi-constraint	Table + clues; fill ends first, then middles.	Clue skipped; messy table.
Row same-facing	Facing North: left = left; South: mirror everything.	Left-always reflex; face unchecked.
Double row	A-left = B-right across facing rows (mirror).	Same-row left applied across.
Unknown/NOT	Blanks + possibilities; NOT/neither eliminates via cross.	Probably-guess; NOT ignored.
Circular center	Center-facing: left = anticlockwise, right = clockwise.	Row rules on a circle.
Circular outside	Reversed: left = clockwise, right = anticlockwise.	Center rules reused; reverse missed.
Mixed facing	Mark in/out arrows per person; track each rule.	All-center assumed; marks missing.
Neighbor/anchor	Immediate = both sides; anchor fixed person, build rest.	Skipped neighbors; no anchor drift.
Venn basics	All A are B: A inside B; No: disjoint; Some: overlap; Some-not: part outside.	Some read as All; No read as opposite.
Syllogism traps	Possibility follows only if possible; Only-A-are-B reverses carefully; two negatives give nothing; use given facts, not real world.	Outside knowledge; middle term dropped.

60-SECOND SPEED ROUND (cover answers, recall aloud)

Q: South-facing, 3rd left of X at pos 5?	A: Mirror: 8th from left
Q: Center-facing, 2nd right of Y (pos 1 of 6)?	A: Clockwise 2 = pos 3
Q: Two negatives conclude?	A: No valid conclusion

Batch 4 (Units VIb-VIIa): Statements + assumptions + data sufficiency + clocks

CONCEPT	FORMULA / RULE	TRAP
General/specific	Match conclusion scope to premise; check distribution.	Scope mixed; undistributed middle.
Two conclusions	Test I, II, both, neither, either-or pairs.	One-true rush; either-or misjudged.
Negation test	Deny the option; if conclusion breaks, it is assumed.	Affirming instead of denying.
Ad assumptions	Best/safest claims hide comparisons; decode the pitch.	Hidden claim missed.
Cause order	Cause precedes effect; rain gives wet, not reverse.	Reversed causation.
Common cause	Ice-cream vs drowning share summer; independent otherwise.	Correlation called causation.
A + R blocks	Verify A true, R true, then R-explains-A link separately.	Both-true assumed to explain.
Sufficiency solo	Test Stmt I alone, then II alone, before combining.	Both-needed reflex; solo never tested.
Together/neither	Combine only if solos fail; nested Venn decides subset Qs.	Partial data called enough.
Three-set Venn	Fill center-out; real-class diagrams picked by matching.	Center filled last; diagram mismatched.
Mirror clock	Mirror flips left-right; 3:00 mirrors 9:00; water flips up-down.	Mirror = water confused.
Hand angles	Angle = $ 30H - 5.5M $; 6:00 gives 180 deg; relative positions per slot.	Hour-only; minute hand ignored.

60-SECOND SPEED ROUND (cover answers, recall aloud)

Q: Mirror image of 5:00?	A: 7:00
Q: Angle at 9:00?	A: 90 deg
Q: Mirror vs water axis?	A: Vertical vs horizontal

Batch 5 (Unit VIIb): Clocks/calendar + images + paper folding + series figures

CONCEPT	FORMULA / RULE	TRAP
Hand angle	$ 30H - 5.5M $; 2:30 gives $ 60 - 137.5 = 77.5$ deg.	Hour-only estimate.
Leap + odd days	Century only if /400 (2100 no); 700 days = 0 odd days.	Century-always; 400-rule missed.
Same calendar	Leap +28, ordinary +11/+6; Jan-Feb count with prior year in leap.	Gap guessed; leap missed.
Mirror/water glyphs	Mirror = L-R flip, water = U-D flip; symmetric A,H,I,M,O,T,0,8 stay.	Axis mixed; asymmetric assumed same.
Clock in mirror	Combine time-mirror with glyph rules.	One rule applied, other dropped.
Single/multiple folds	Mirror hole across each fold line; layers multiply holes.	Hole stays put; unfold skipped.
Cuts + direction	Track every fold, then unfold corner/edge cuts in reverse.	Direction untracked; shape guessed.
Rotation/movement	90/180 clockwise steps; N-to-3x90-clockwise ends West.	Degrees uncounted; direction flipped.
Shading/count	Fill-pattern + add/remove (+1/-1) element rules.	Fill-blind; count skipped.
3x3 matrix	Row AND column rules; shape+shade+count combine.	Row-only; traits dropped.

60-SECOND SPEED ROUND (cover answers, recall aloud)

Q: Angle at 2:30?	A: 77.5 deg
Q: Leap year 2100?	A: No (century/400)
Q: Mirror-symmetric capitals (3)?	A: A, H, I

Batch 6 (Unit VIII): Figure counting + dice/cubes + puzzle strategy

CONCEPT	FORMULA / RULE	TRAP
Outline vs flip	Trace perimeter; rotation keeps handedness, reflection mirrors (R stays R on turn).	Turn/flip mixed.
Triangle count	Sweep systematically left-right-top-down + formula check.	Random glance; formula skipped.
Squares 4x4	$16+9+4+1 = 30$; n^2 + sum pattern.	Eyeball count.
Rectangles	LxW choice product; verify via grid formula.	Formula missed.
Cube net/dice	Opposites never adjacent; common-face rotation finds opposite.	Adjacent guessed as opposite.
Standard die	Opposite faces sum 7 (1-6, 2-5, 3-4).	Rule unchecked.
Painted cube	Corners = 8 three-face; edges $12 \times (n-2)$ two-face; $(n-2)^3$ zero-face (n5 gives 36 two-face).	Visible-only count; formula skipped.
Partial paint	Paint-then-cut vs cut-then-paint differ; diagram first.	Imagined only; cut order ignored.
Clue priority	Definite + most-constrained clues first; anchor A then build.	Weak-first drift; stuck too long.
Restriction words	only/never/exactly constrain hard; recheck every clue ticked.	Probably-reading; box unjustified.
Bailout + log	30-45s stuck: mark, move, log error class (misread/calc/rush/guess).	One-more-minute death; no review.

60-SECOND SPEED ROUND (cover answers, recall aloud)

Q: Squares in 4x4 grid?	A: 30
Q: n5 cube, 2-face count?	A: $12 \times 3 = 36$
Q: Standard die: 1 opposite?	A: 6